Rajveer Gupta

+91-8563938645 • rajveercareer1@gmail.com • LinkedIn • GitHub • Kaggle • Hugging Face • Portfolio

Research Interests

Where **computer vision meets language**: vision-language models, multimodal retrieval, and efficient Transformers and state-space models, tested on real-world problems against honest baselines with ablations.

Education

Indian Institute of Information Technology, Nagpur Aug 2024 – Jun 2028
B.Tech, Computer Science & Engineering — CGPA: 9.08/10 *Nagpur, India*
Coursework: DSA, OOP (AA), Software Engineering (AA), Operating Systems (AB), Programming Language Design (AB), DBMS

Publications

A. Gupta, A. Singh, **R. Gupta**, P. Jain, T. Patidar, "Fault Tolerant Industrial Vision Framework Through Semantic-Guided Cross-View Temporal Reconstruction and Restoration," *IEEE IRAI 2026*, Melbourne, Australia (accepted and presented).

Research Experience

Research Intern, Computer Vision — IIIT Nagpur (Advisor: Dr. Nishat Afshan Ansari) 2026
What it does: when a factory CCTV camera fails, our framework detects it and rebuilds the missing view from nearby cameras.

- Co-developed a four-stage pipeline: **hysteresis-based fault classification** that flags a camera only when a fault persists, pose-free **3D Gaussian cross-view reconstruction** (NoPoSplat), semantic-guided **diffusion** to fill unseen regions, and a **Mamba-CNN** restorer with linear-time sequence modeling instead of quadratic attention.
- Reached **30.02 dB PSNR / 0.893 SSIM** on Wildtrack and **32.42 dB / 0.953 SSIM** on GoPro, outperforming NAFNet and MoCE-IR; ran ablations isolating each module's contribution.

Research Projects

Resolv — Multilingual Entity Resolution over 11.7M Records (Amazon ML Challenge 2026) 2026
GitHub — Team lead of 4 · AIR 471 of 32K teams · PyTorch, multilingual-e5, Qwen3-Reranker, LightGBM, Polars
What it does: finds every record of the same real business across French, Indian and US databases, despite different spellings and scripts.

- **Diagnosed a train–test distribution shift** (39.8% vs 26.0% unmatched records) behind a 0.963 → 0.94 drop from validation to leaderboard; simulated it by turning 20% of training businesses into noise and refit all models on it.
- Paired IDF-weighted blocking keys with a multilingual-e5 bi-encoder fine-tuned with **InfoNCE**: **99.86% pair recall** vs 95.75% for keys alone; a LightGBM re-ranker cut 86.7 candidates per business to 5.58.
- Stacked 107 hand-built features with an e5 cross-encoder and a fine-tuned **Qwen3-Reranker-0.6B** run only on uncertain pairs; ablations traced every gain, lifting leaderboard F0.5 from 0.94 to **0.986** (0.998 validation precision).

ReturnGuard — Vision-Language Return-Fraud Detection 2026
GitHub / Demo — Python, LangGraph, DINOv2, Gemini VLM, Flask, SQLite
What it does: checks whether a returned product is really the one that was delivered, catching swaps or damage before the refund.

- Built a two-tier pipeline: **DINOv2** patch-mean embeddings auto-clear near-identical items, and a reusable **VLM subgraph** reviews the rest in parallel per photo view before an LLM judge assigns a 0–100 risk score.
- Found that DINOv2 rated *different* same-material products 85–92% similar, tightened the threshold and added sharpness weighting for blurry photos: **0.93 precision / 0.88 F1** on 36+ hand-labelled pairs.

Transformers from First Principles & Parameter-Efficient Fine-Tuning Study 2025 – 2026
BERT / ViT / PEFT (Kaggle) — PyTorch, TensorFlow, Hugging Face PEFT
What it does: rebuilds core AI models by hand, then measures which fine-tuning method gives the most accuracy per unit of compute.

- Pre-trained a **BERT-style encoder from random weights** (MLM + NSP, custom tokenizer); wrote a GPT-style decoder and a **Vision Transformer** from scratch (81% on chest X-rays).
- Benchmarked cross-domain sentiment transfer (AG News → 50K IMDB): full fine-tuning 65%, Adapters 78%, LoRA 85%, **QLoRA 86% with 75% less GPU memory**.

Memora — Face Recognition and Speech-Distress Detection for Alzheimer's Care 2025 – 2026
GitHub / Demo — DeepFace (ArcFace), BERT · Team lead; I built the ML layer

- Combined ArcFace face recognition (**98.5% on LFW**) with BERT speech-distress detection; stores embeddings, not faces.

Achievements

- **AIR 471 of 32K teams**, Amazon ML Challenge 2026 • **Amazon ML Summer School 2026**, top 3,000 of 134,000+.
- **Best Innovation Award**, IIT Hyderabad • Imagine Cup 2025 MVP Round • IEEE IES GenAI Hackathon Finalist.

Skills

Languages: Python, C++, C, Java, SQL
Machine Learning: PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, PEFT (LoRA / QLoRA), LightGBM
Vision & Language: ViT, DINOv2, VLMs, 3D Gaussian Splatting, Diffusion, Mamba / SSMs, multilingual-e5, OpenCV
Tools: LangGraph, LangChain, FAISS, ChromaDB, Polars, Flask, FastAPI, Docker, Git, Linux